UNIT 1: Computer Architecture Concepts & ARM – Question Bank with Bloom's Taxonomy

✅ PART-A Questions (2 Marks Each – 15 Questions)

(*Cognitive levels: Remember (R), Understand (U), Apply (A)*)

1. Define RISC and CISC. *(R)*
2. What is meant by instruction set architecture? *(U)*
3. Mention any two features of RISC architecture. *(U)*
4. State two differences between RISC and CISC. *(U)*
5. What are the operations and operands in instructions? *(R)*
6. Define the term "addressing mode." *(R)*
7. What is the use of logical operations in processors? *(U)*
8. What do you mean by "Instruction Performance"? *(U)*
9. Write the Classic CPU Performance Equation. *(R)*
10. Define the term "Power Wall" in processor design. *(U)*
11. List the technologies used to build processors. *(R)*
12. List the technologies used to build memory. *(R)*
13. What are the eight ideas of computer architecture? *(R)*
14. Name any two instruction types in ARM. *(U)*
15. What is the role of operands in instruction execution? *(U)*

✅ PART-B Questions (13 Marks Each – 10 Questions)

(*Cognitive levels: Apply (A), Analyze (An), Evaluate (E)*)

1. Compare and contrast RISC and CISC architectures in terms of performance, complexity, and power consumption. *(An)*
2. Explain the eight great ideas of computer architecture in detail. *(U)*
3. Discuss in detail the technologies used for building processors and memory with suitable examples. *(U)*
4. Derive and explain the Classic CPU Performance Equation. Illustrate with an example. *(A)*
5. Analyze the factors affecting CPU performance. *(An)*
6. Describe various instruction types with examples of operations and operands. *(U)*
7. Explain different types of addressing modes with examples. *(A)*
8. Discuss the concept of power wall and its implications in modern processors. *(E)*
9. Analyze instruction-level performance and show how performance is measured. *(An)*
10. Explain the logical operations and instructions used for decision-making in computer architecture. *(U)*

✅ PART-C Questions (15 Marks Each – 2 Questions)

(*Cognitive levels: Create (C), Evaluate (E), Analyze (An)*)

1. **Case Study: ARM Architecture**
   Analyze the ARM processor architecture in detail. Explain how it supports RISC principles. Highlight instruction format, pipeline stages, and addressing modes with examples. *(An, E)*
2. **Design-Based Problem**
   Design a hypothetical CPU with 5 instructions using RISC principles. Define instruction format, logical operations, and addressing modes. Evaluate its performance using the classic CPU performance equation with given instruction count, CPI, and clock rate. *(C, E)*

UNIT 2: ALU Design & Arithmetic Operations

(With Bloom's Taxonomy Categorization)

✅ PART-A Questions (2 Marks Each – 15 Questions)

(*Cognitive levels: Remember (R), Understand (U), Apply (A)*)

1. Define ALU. *(R)*
2. What are the components of an ALU? *(U)*
3. What is the difference between signed and unsigned addition? *(U)*
4. State the significance of overflow in arithmetic operations. *(U)*
5. What is sequential multiplication? *(R)*
6. Define Booth's algorithm. *(R)*
7. What is meant by recoding technique in multiplication? *(U)*
8. Mention any two advantages of Booth's algorithm. *(U)*
9. What is bit-pair recoding? *(U)*
10. Differentiate between restoring and non-restoring division. *(U)*
11. Define normalization in floating-point representation. *(R)*
12. What is IEEE 754 floating point representation? *(U)*
13. Write the steps involved in floating point addition. *(A)*
14. Mention one application where floating point arithmetic is preferred. *(U)*
15. What is the role of sign, exponent, and mantissa in floating point numbers? *(U)*

✅ PART-B Questions (13 Marks Each – 10 Questions)

(*Cognitive levels: Apply (A), Analyze (An), Evaluate (E)*)

1. Design a basic ALU block diagram and explain how it performs arithmetic and logic operations. *(A)*
2. Explain the addition and subtraction of signed integers using 2's complement representation with examples. *(A)*
3. Describe the sequential multiplication algorithm with an example. *(A)*
4. Explain Booth's algorithm for multiplication. Illustrate with a 4-bit example. *(A)*
5. Describe the recoding technique and bit-pair recoding used for faster multiplication. *(U, A)*
6. Explain restoring division algorithm with step-by-step example. *(A)*
7. Describe the non-restoring division algorithm and explain how it differs from restoring method. *(A)*
8. Explain the representation of floating-point numbers using IEEE 754 single and double precision formats. *(U)*
9. Illustrate the steps of floating-point addition and subtraction with normalized values. *(A)*
10. Explain floating-point multiplication with a worked-out example and mention normalization and rounding steps. *(A)*

✅ PART-C Questions (15 Marks Each – 2 Questions)

(*Cognitive levels: Analyze (An), Create (C), Evaluate (E)*)

1. **Case Study: Design Problem**
   Design and implement an ALU supporting addition, subtraction, and multiplication (using Booth's algorithm). Explain the architecture with block diagram and control logic. *(C, An)*
2. **Application Problem: Floating Point Evaluation**
   A processor needs to handle floating point arithmetic operations in a scientific computation module. Propose and evaluate the most suitable method for floating point multiplication and division, highlighting accuracy, rounding, and exception handling. *(E)*

UNIT 3: MIPS Implementation and Pipelining

✅ PART-A: 2 Marks Questions (15 Questions)

(*Cognitive Levels: Remember (R), Understand (U), Apply (A)*)

1. What is a datapath in processor design? *(R)*
2. Define control unit in a processor. *(R)*
3. What is a single-cycle datapath? *(U)*
4. Name the five stages of instruction execution in pipelining. *(R)*
5. Define pipelining in computer architecture. *(U)*
6. List any two advantages of pipelining. *(U)*

7. Mention any two disadvantages of pipelining. *(U)*
8. What are data hazards? *(R)*
9. List the three types of data hazards. *(R)*
10. Define control hazard. *(R)*
11. What is branch prediction? *(U)*
12. What is an exception in a pipelined processor? *(U)*
13. State one method to resolve data hazards. *(U)*
14. Mention one technique to resolve control hazards. *(U)*
15. What is forwarding (bypassing)? *(U)*

✅ PART-B: 13 Marks Questions (10 Questions)

(*Cognitive Levels: Apply (A), Analyze (An), Evaluate (E)*)

1. Explain the construction of a single-cycle datapath for MIPS processor with a neat diagram. *(A)*
2. Discuss the working of a control unit for a simple datapath. Include signal generation and execution. *(A)*
3. Describe in detail the five stages of a pipelined MIPS architecture. *(U, A)*
4. Explain pipelined datapath with a suitable diagram and compare it with non-pipelined execution. *(A, An)*
5. What are pipeline hazards? Explain the types of data hazards with examples. *(U, An)*
6. Describe the techniques for minimizing data hazards such as forwarding and stall. *(A, E)*
7. Explain control hazards with examples and compare different schemes to resolve them. *(A, E)*
8. Discuss static and dynamic branch prediction algorithms. *(U, An)*
9. What are exceptions in pipelining? How are they handled in MIPS architecture? *(U, A)*
10. Evaluate the performance benefits and challenges of pipelining in modern processors. *(E)*

✅ PART-C: 15 Marks Questions (2 Questions)

(*Cognitive Levels: Analyze (An), Create (C), Evaluate (E)*)

1. **Case Study: Design Challenge**
   Design a pipelined datapath for MIPS including forwarding and hazard detection logic. Explain how the pipeline handles instruction execution with and without hazards. *(C, An)*
2. **Application Problem: Branch Prediction**
   Compare and evaluate different branch prediction algorithms (static and dynamic) in terms of prediction accuracy and execution speed for a loop-heavy instruction set. Propose an optimized solution. *(E, An)*

UNIT 4: Instruction-Level Parallelism and Advanced Processor Architectures

✅ PART-A: 2 Marks Questions (15 Questions)

(*Bloom's Levels: Remember (R), Understand (U), Apply (A)*)

1. What is instruction-level parallelism (ILP)? *(R)*
2. Define superscalar architecture. *(R)*
3. What are the two main policies used in instruction issue? *(U)*
4. What is register renaming and why is it used? *(U)*
5. Differentiate between in-order and out-of-order execution. *(U)*
6. Define Very Long Instruction Word (VLIW) processor. *(R)*
7. How does the compiler help in VLIW architecture? *(U)*
8. State one key difference between Superscalar and VLIW processors. *(U)*
9. What is Flynn's classification? *(R)*
10. Define SIMD and MIMD with examples. *(U)*
11. What is a shared memory multiprocessor? *(R)*
12. List any two difficulties in writing parallel programs. *(U)*
13. What is meant by dependency checking in ILP? *(U)*
14. What is the purpose of a scoreboard in a processor? *(U)*
15. Define graphics processors in brief. *(U)*

✅ PART-B: 13 Marks Questions (10 Questions)

(*Bloom's Levels: Apply (A), Analyze (An), Evaluate (E)*)

1. Explain instruction-level parallelism with examples. How is ILP exploited in modern processors? *(A)*
2. Describe superscalar processor architecture and discuss the instruction issue policies. *(A)*
3. What is register renaming? Explain how it resolves data hazards with an example. *(A, An)*
4. Illustrate the implementation of a superscalar processor with instruction flow diagram. *(A)*
5. Describe the working of VLIW processors. How does the compiler aid in parallelism? *(U, A)*
6. Compare Superscalar and VLIW processors in terms of performance, complexity, and control. *(An)*
7. Explain Flynn's classification with examples for each category (SISD, SIMD, MISD, MIMD). *(U, A)*
8. Discuss the major difficulties in designing and developing parallel programs. *(An, E)*
9. Explain shared memory multiprocessors and how they enable parallel computation. *(U, A)*

10. Write a short note on graphics processors and their role in high-performance computing. *(U, A)*

✅ PART-C: 15 Marks Questions (2 Questions)

(*Bloom's Levels: Analyze (An), Create (C), Evaluate (E)*)

1. **Case Study: AMD Ryzen Performance Optimization**
   Analyze the microarchitectural features of AMD Ryzen processors. How do instruction-level parallelism, caching techniques, and multi-threading contribute to performance? *(An, E)*
2. **Design-Based Question**
   Propose an optimized VLIW architecture for a media processing application. Define instruction formats, scheduling strategy, and compare with a superscalar approach in terms of execution efficiency. *(C, E)*

UNIT 5: Memory Hierarchy, Virtual Memory, I/O System & Multicore Smartphones

✅ PART-A: 2 Marks Questions (15 Questions)

(*Bloom's Levels: Remember (R), Understand (U), Apply (A)*)

1. What is memory hierarchy? *(R)*
2. List two commonly used memory technologies. *(R)*
3. What is cache memory and why is it needed? *(U)*
4. Define direct-mapped cache. *(R)*
5. Differentiate between fully associative and set-associative mapping. *(U)*
6. What is the role of replacement algorithms in cache memory? *(U)*
7. What is multi-level cache? *(U)*
8. Define virtual memory. *(R)*
9. What is paging in virtual memory? *(U)*
10. Define segmentation. *(R)*
11. What is the purpose of the Translation Look-aside Buffer (TLB)? *(U)*
12. What is the difference between physical and virtual addresses? *(U)*
13. What is programmed I/O? *(R)*
14. Differentiate between DMA and interrupt-driven I/O. *(U)*
15. What is an I/O processor? *(U)*

✅ PART-B: 13 Marks Questions (10 Questions)